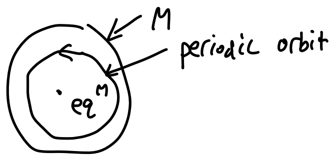
$\boxed{\text{2026 Jan 29}}$

Recall PB Theorem: (planar)

If $\exists$ positively invariant compact set M with no equilibrium

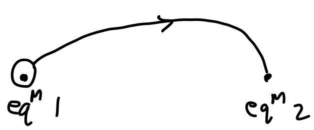
then $\exists$ a periodic orbit in M



General form of PB theorem:

For (time invariant, planar systems), bounded trajectories converge to equilibria, periodic orbits or union of equilibria connected by trajectories

Ex. equilibria connected by trajectories



No "chaos" in 2D time-invariant continuous time system.

vs. **DT:** $x_{t+1} = f(x_t)$

CT: $\dot{x} = f(x)$

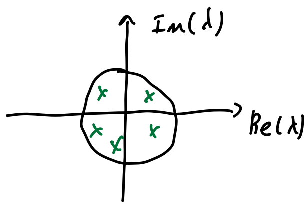
"fixed points": $f(x) = x$

Asymptotic stability criterion $\quad x_{t+1} = A x_t$

$|\lambda_i(A)| < 1 \quad \forall i \qquad A = \frac{\partial f}{\partial x}\Big|_{x=0}$

$\lambda_i = a_i + j b_i$

$|\lambda_i| = \sqrt{a_i^2 + b_i^2}$



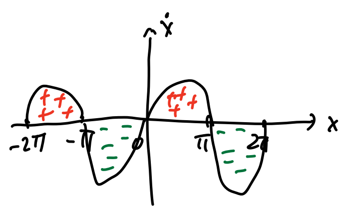
If $|\lambda_i(A)| > 1$ for some i $\Rightarrow$ $x = 0$ is unstable

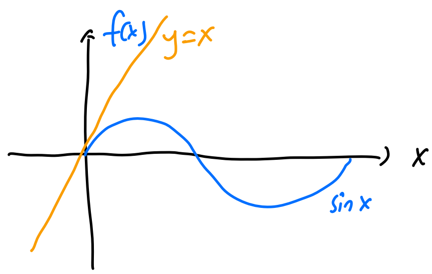
Cobweb Diagram

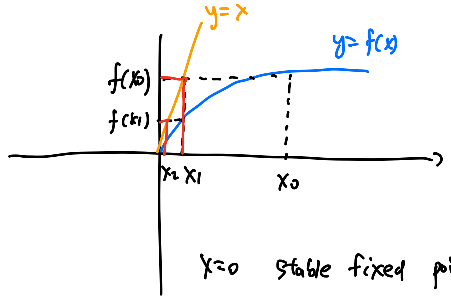
$x_{t+1} = \sin(x_t)$ $\qquad \dot{x} = \sin x$

fixed point: $\sin x = x$

$\Rightarrow x = 0$







$x_1 = f(x_0)$

$x_2 = f(x_1)$

$x = 0$ stable fixed point

$f(x) = \sin x$

$\frac{\partial f}{\partial x} = \cos x$

$\left.\frac{\partial f}{\partial x}\right|_{x=0} = \cos 0 = 1$ $\qquad x_1 = f(x_0)$

$x_2 = f(x_1)$

Linearization cannot evaluate stability $x=0$ for $x_{t+1} = \sin(x_t)$

<u>D1</u> Periodic orbit has period T

if $x_{t+T} = x_t \quad \forall t$

$x_6 = x_4 = x_2 = x_0 \qquad x_{t+1} = f(x_t)$

$x_7 = x_5 = x_3 = x_1 = f(x_0)$

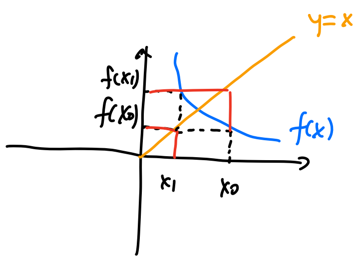
$x_2 = f(x_1)$

$x_2 = f(f(x_0))$

$x_1 = f(f(x_1))$

Claim : if x_0 is a fixed point of $f \circ f$

x_1 a fixed point of $f \circ f$



<u>Ex.</u> $x_{t+1} = r(1-x_t)x_t \qquad 4 > r > r$

<u>logistic growth model</u>

$\max_{x\in[0,1]} (1-x)x = \frac{1}{4}$ at $x = \frac{1}{2}$

$x_0 \in [0,1] \Rightarrow x_t \in [0,1] \quad \forall t$

Fixed points: $f(x) = r(1-x)x$

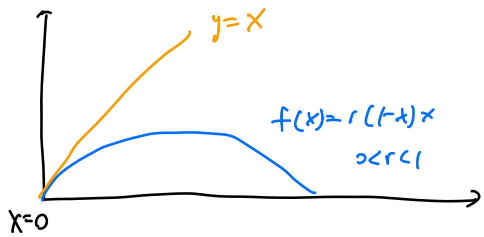
$$f(x) = x$$

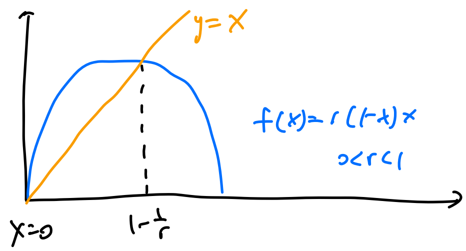
$$r(1-x)x = x$$

$$\Downarrow$$

$x=0$ $\quad r(1-x) = 1$

$x = 1 - \frac{1}{r} \quad r>1$





$$\frac{df}{dx} = \frac{d}{dx} r(1-x)x$$

$$= r(1-2x)$$

$$\frac{df}{dx}\Big|_{x=0} = r$$

$$\frac{df}{dx}\Big|_{x=1-\frac{1}{r}} = r\left[1-2\left(1-\tfrac{1}{r}\right)\right] = r\left[-1+\tfrac{2}{r}\right] = -r+2$$

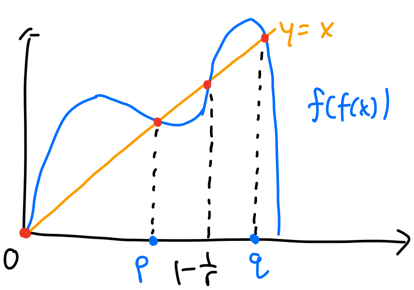
$$f \circ f(x) = f(rx(1-x))$$

$$= ry(1-y) \qquad y = rx(1-x)$$

$$= r \cdot rx(1-x) \cdot [1-rx(1-x)]$$

$$= r^2 x(1-x) \cdot [1-rx+rx^2]$$

$\boxed{r=3}$



Periodic orbit : $p \longleftrightarrow q$

$r_1 = 3$: period-2 cycle born

$r = 3.4494$: period-4 cycle born

$r = 3.544$: period 8 cycle born

⋮

$r_\infty = 3.5699$: Chaotic behavior

⋮

$r = 3.83$: come out of chaotic behavior

<u>Ex.</u>

$$\dot{x}_1 = x_1 x_2 \qquad = f_1(x_1, x_2)$$

$$\dot{x}_2 = -x_2 - x_1^2 \qquad = f_2(x_1, x_2)$$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} x_2 & x_1 \\ -2x_1 & -1 \end{bmatrix}$$